Analyze E-Commerce Sales to Improve Customer Conversion and Retention Using Power-bi

1.Abstract:

The rapid growth of e-commerce has generated enormous volumes of transactional and customer data. Organizations that effectively analyze this data can improve customer acquisition, retention, profitability, and operational efficiency. The purpose of this project was to develop an interactive Power BI dashboard capable of providing actionable insights into sales performance, customer behavior, product profitability, and customer experience. The project involved data cleaning, transformation, modeling, KPI creation, and dashboard development using Power BI. Multiple datasets related to customers, orders, products, reviews, and geography were integrated into a unified analytical model. The final dashboard enables stakeholders to monitor business performance through
intuitive visualizations and KPI indicators. The completed dashboard consists of four analytical pages: Home, Sales & Product Analysis, Customer Analysis, and Orders & Customer Experience. These pages collectively provide a comprehensive understanding of the factors influencing revenue growth, profitability, customer retention, and service quality.

2.Introduction:

In today's competitive digital marketplace, businesses rely heavily on data-driven decision-making. E-commerce platforms collect information regarding customer purchases, payment methods, product performance, order full-fillment, cancellations, and customer reviews.

However, raw transactional data alone cannot support strategic decision-making. Therefore, it becomes essential to transform this data into meaningful insights through data analytics and business intelligence tools.

Power BI was selected for this project due to its powerful visualization capabilities, data modelling features, and ability to create interactive dashboards. The dashboard developed in this project helps management identify high-performing products, understand customer purchasing patterns, evaluate operational efficiency, and improve customer satisfaction.

3. Problem Statement:

The business lacked a centralized reporting system capable of monitoring key performance indicators across sales, customers, products, and operations.

Without a consolidated analytical dashboard:

- Revenue trends remain difficult to monitor.
- Product profitability cannot be easily evaluated.
- Customer behaviour patterns remain hidden.
- Cancellation and return trends are difficult to identify.
- Customer satisfaction drivers remain unclear.
- Therefore, there was a need to develop a Power BI dashboard capable of transforming raw business data into actionable insights.

4. Business Objectives:

The primary objectives of this project were:

- Objective 1

To identify category- and segment-level drivers of first purchase conversion and overall sales growth.

- Objective 2

To understand repeat purchase, volume, and loyalty patterns across demographics, categories, and delivery performance.

- Objective 3

To optimize product portfolio, pricing, discounting, and shipping for profitable growth.

- Objective 4

To leverage reviews, ratings, and customer sentiment to enhance conversion and retention.

- Objective 5

To evaluate logistics and operational efficiency to improve customer experience.

- Objective 6

To optimize customer retention and lifetime value across key product categories.

5. Datasets Used:

The project utilized multiple e-commerce datasets.

1. Customer Dataset

Contains customer identification and segmentation information.

2. Orders Dataset

Contains order details including order status and timestamps.

3. Order Items Dataset

Contains product-level transaction information.

4. Product Dataset

Contains product category and product information.

5. Reviews Dataset

Contains customer ratings and review scores.

6. Geolocation Dataset

Contains location-related information for customer analysis.

These datasets were integrated to create a complete analytical model.

6. Data Cleaning and Preparation

Before building the dashboard, extensive data cleaning was performed.

Step 1: Import Data

All datasets were imported into Power BI.

Step 2: Data Type Validation

Date, numeric, and text columns were validated.

Step 3: Duplicate Removal

Duplicate records were identified and removed.

Step 4: Missing Value Handling

Null values were identified and treated appropriately.

Step 5: Standardization

Text fields were standardized for consistency.

Step 6: Relationship Validation

Primary and foreign key relationships were validated.

Step 7: Date Table Creation

A dedicated date dimension was created.

Step 8: Customer Segmentation

Customers were classified into:

- New
- Returning
- Loyal

Step 9: Calculated Columns Creation

Additional calculated columns were created such as:

- Revenue
- Profit
- Delivery Delay
- Cancellation Rate
- Return Rate

Step 10: Data Model Optimization

The final model was optimized for performance and reporting.

7. Data Modelling

A star schema model was implemented.

Dimension Tables

- DimCustomer
- DimDate
- DimGeography
- Products

Fact Tables

- FactOrders
- FactSales
- FactReviews

Relationships were created between dimension tables and fact tables to support efficient analysis and filtering.

This model allowed users to drill down into detailed insights while maintaining fast dashboard performance.

8. KPI Development

The following KPIs were developed.

- Total Revenue

Measures total business revenue generated.

- Total Profit

Measures overall profitability.

- Profit Margin %

Measures profitability efficiency.

- Total Orders

Measures total completed transactions.

- Total Customers

Measures unique customers.

- Average Order Value

Measures average spending per order.

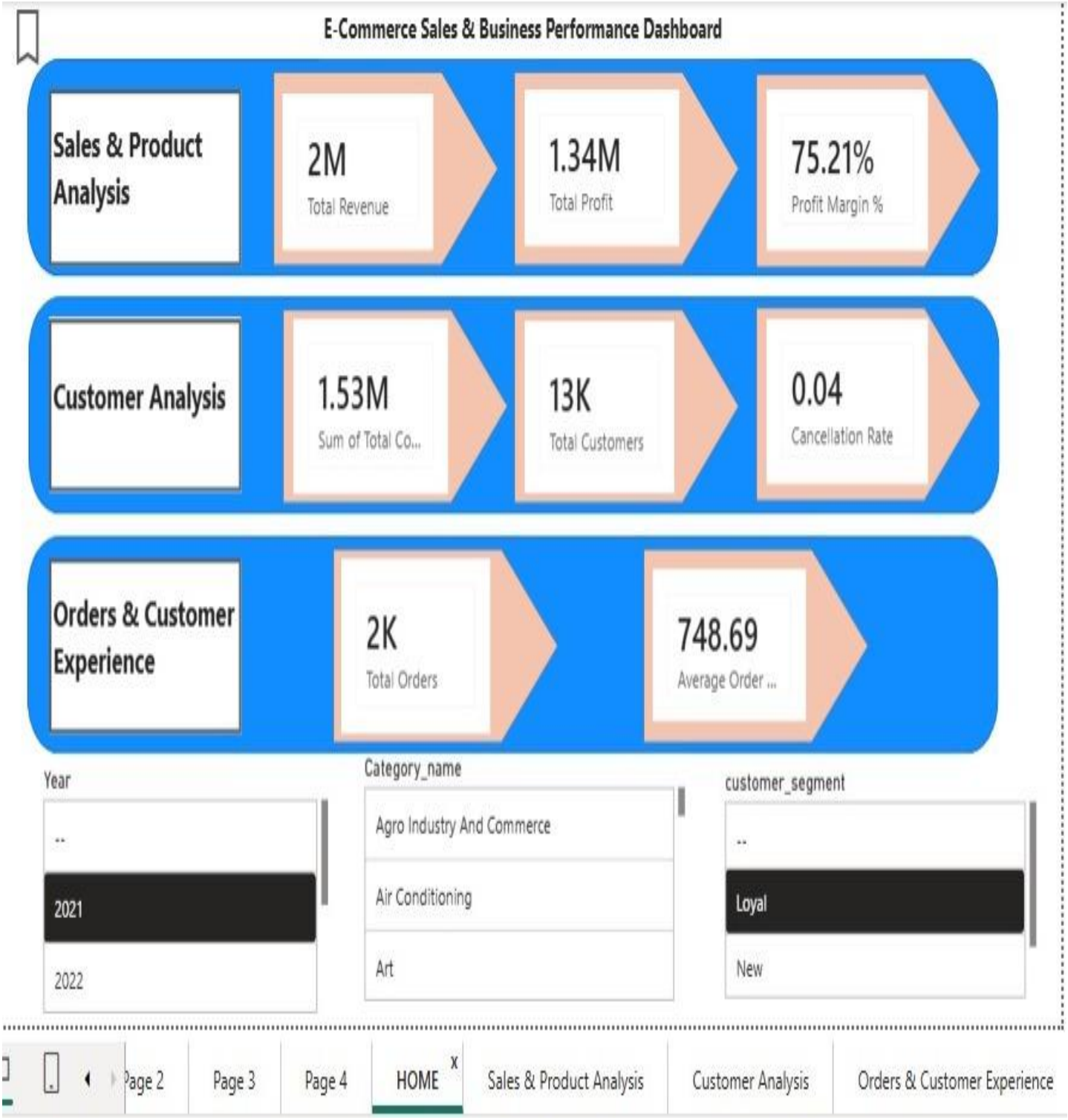
- Cancellation Rate

Measures percentage of cancelled orders.

These KPIs appear on the Home dashboard page and provide an executive-level business summary.

9. Dashboard Design and Analysis

PAGE 1 – HOME DASHBOARD:



Purpose

The Home page serves as the executive summary page.

KPIs Displayed

- Total Revenue = 2M
- Total Profit = 1.34M
- Profit Margin = 75.21%
- Total Customers = 13K
- Total Orders = 2K
- Average Order Value = 748.69
- Cancellation Rate = 0.04

Filters Used

- Year
- Product Category
- Customer Segment

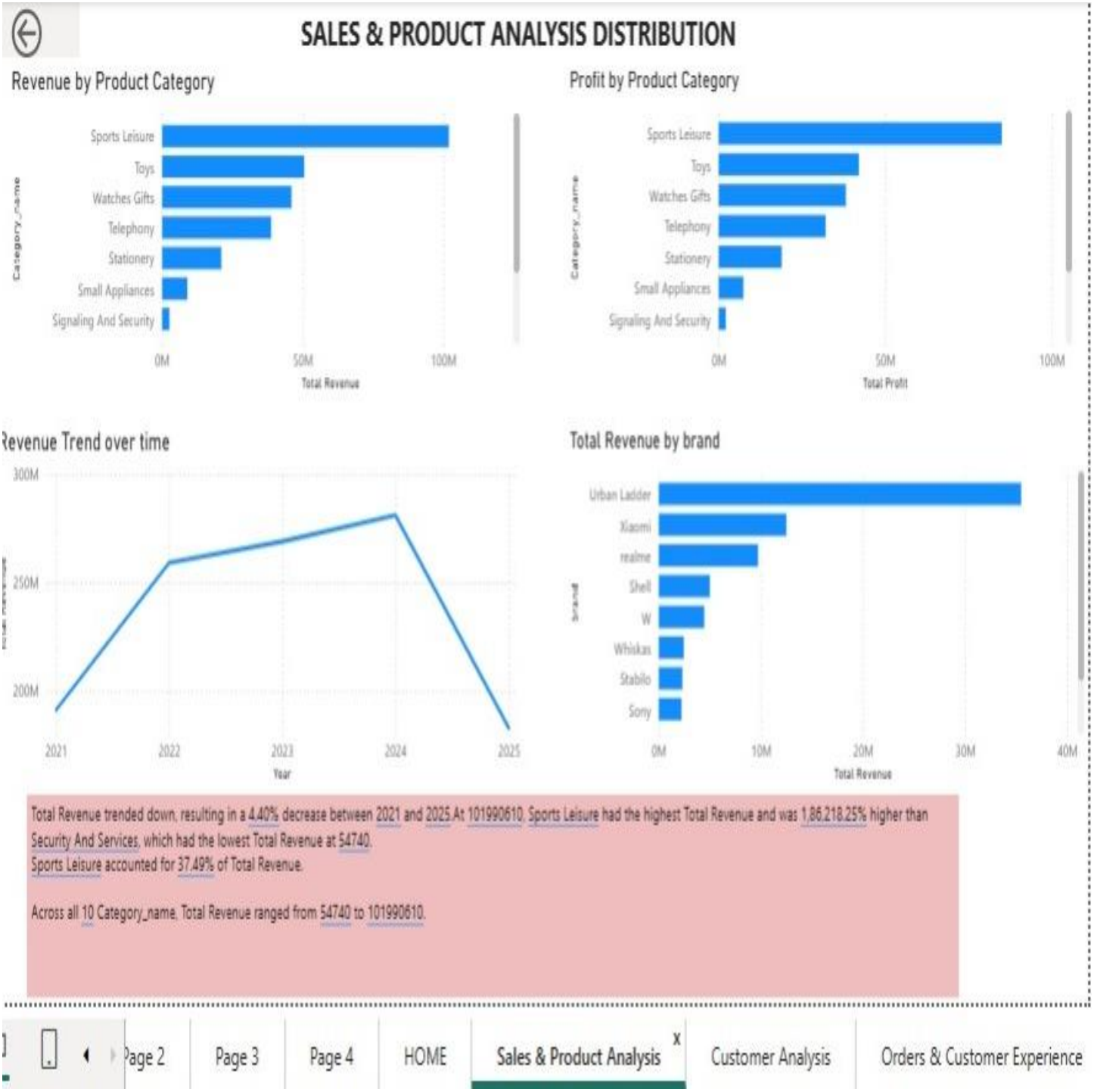
Business Value

This page provides management with a quick overview of business performance without requiring detailed exploration.

Interpretation

The business generated strong profitability, as reflected by the profit margin exceeding 75%. Customer and order volumes indicate healthy business activity while the low cancellation rate suggests stable operational performance.

PAGE 2 – SALES & PRODUCT ANALYSIS:



Purpose

This page evaluates product performance and revenue generation.

Visual 1: Revenue by Product Category

Shows revenue contribution by category.

Finding

Sports Leisure generated the highest revenue among all product categories.

Visual 2: Profit by Product Category

Shows profitability by category.

Finding

Sports Leisure also generated the highest profit.

Visual 3: Revenue Trend Over Time

Displays yearly revenue movement.

Finding

Revenue increased steadily until 2024 before experiencing a decline in 2025.

Visual 4: Revenue by Brand

Displays top-performing brands.

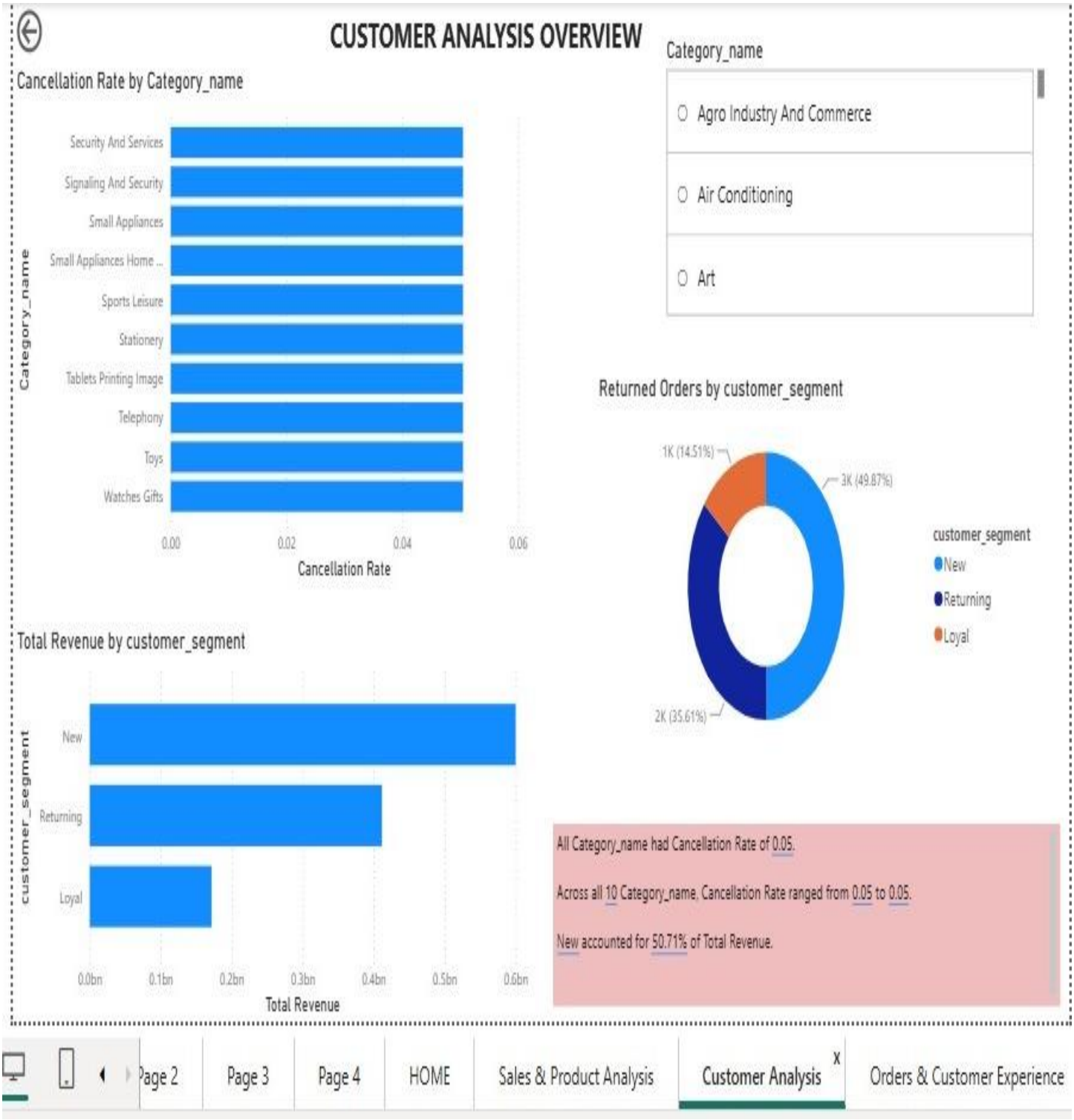
Finding

Urban Ladder emerged as the highest revenue-generating brand.

Business Insight

The organization should prioritize investment in high-performing categories and brands while investigating declining revenue trends.

PAGE 3 – CUSTOMER ANALYSIS:



Purpose

This page focuses on customer behaviour and retention.

Visual 1: Cancellation Rate by Category

Displays cancellation percentages across categories.

Finding

Cancellation rates remain relatively consistent across categories.

Visual 2: Revenue by Customer Segment

Compares revenue generated by customer groups.

Finding

New customers generated the highest revenue contribution.

Visual 3: Returned Orders by Customer Segment

Displays returned orders by segment.

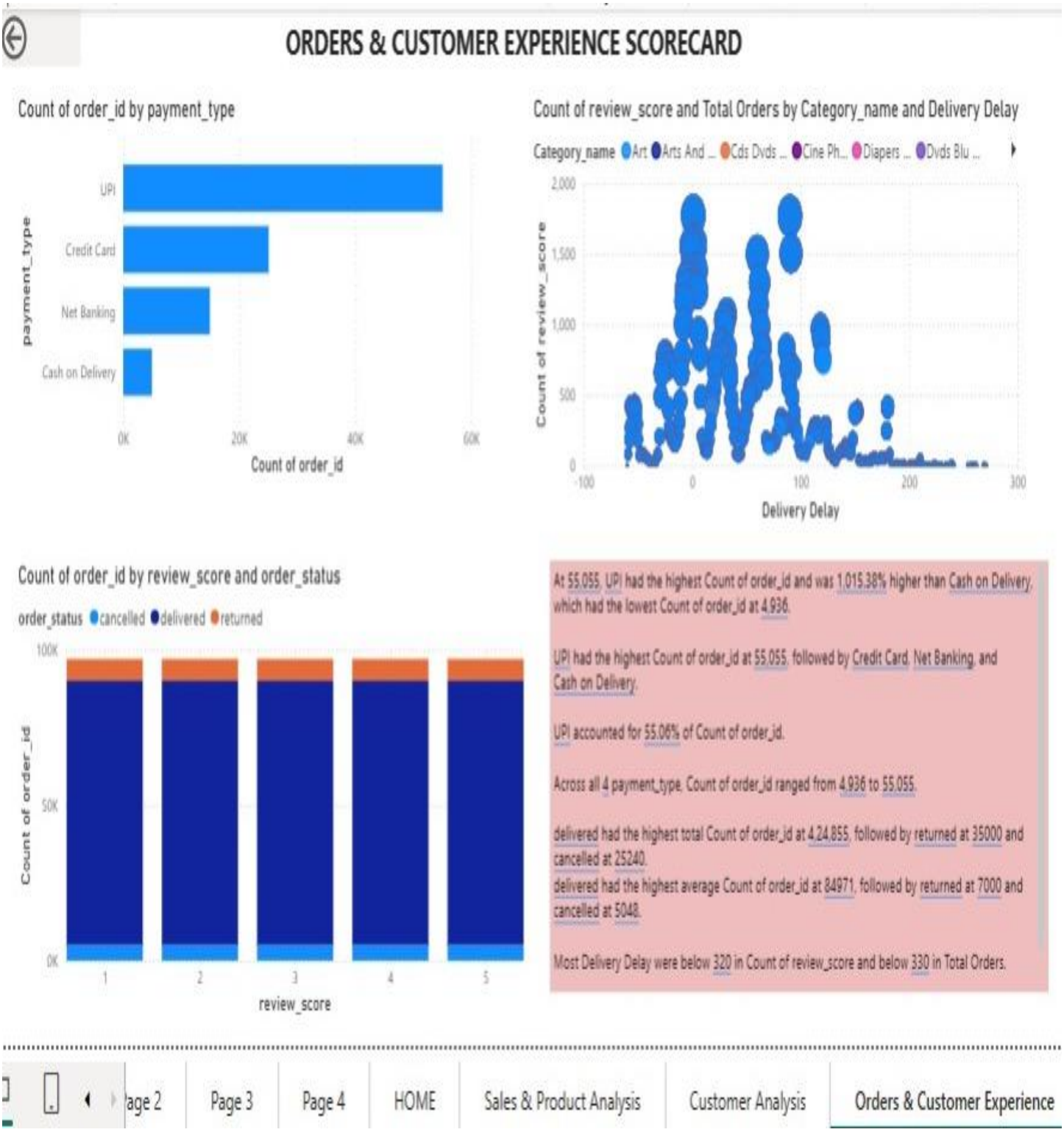
Finding

Returning customers contributed significantly to return activity.

Business Insight

Customer acquisition remains strong, but retention strategies should be enhanced to convert new customers into loyal customers.

PAGE 4 – ORDERS & CUSTOMER EXPERIENCE:



Purpose

This page evaluates operational performance and customer satisfaction.

Visual 1: Orders by Payment Type

Displays preferred payment methods.

Finding

UPI is the most preferred payment method.

Visual 2: Delivery Delay vs Review Score

Scatter plot showing relationship between delays and customer reviews.

Finding

Orders with lower delivery delays tend to receive better customer ratings.

Visual 3: Review Score vs Order Status

Shows review score distribution across order statuses.

Finding

Delivered orders dominate across all review scores.

Business Insight

Improving delivery performance can directly improve customer satisfaction and review scores.

10. Key Findings:

The analysis revealed several important insights:

- Sports Leisure is the strongest revenue-generating category.
- Urban Ladder is the top-performing brand.
- UPI is the most widely used payment method.
- New customers contribute the highest revenue.
- Delivery performance influences customer satisfaction.
- Low cancellation rates indicate operational stability.
- Customer segmentation provides opportunities for retention-focused marketing strategies.

11. Recommendations:

Based on the analysis, the following recommendations are proposed:

- Recommendation 1

Increase inventory and promotional efforts for high-performing categories.

- Recommendation 2

Develop retention programs targeting new customers.

- Recommendation 3

Investigate causes of declining revenue after 2024.

- Recommendation 4

Improve logistics processes to reduce delivery delays.

- Recommendation 5

Introduce loyalty programs for repeat customers.

- Recommendation 6

Continue expanding UPI-based payment incentives due to customer preference.

12. Conclusion:

This project successfully transformed raw e-commerce data into a meaningful business intelligence solution using Power BI. Through data cleaning, modelling, KPI development, and dashboard design, valuable insights were generated regarding sales performance, product profitability, customer behaviour, and customer experience.

The dashboard enables stakeholders to monitor business performance in real time and supports informed decision-making. The findings and recommendations derived from the analysis can assist management in improving customer conversion, increasing retention, optimizing product strategy, and enhancing operational efficiency.